



UTM
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Final Project Report (WorkBeacon)

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1. Stakeholders and Users Identification

1.1 Stakeholder Overview

The primary stakeholder of the WorkBeacon project is the Human Resources (HR) and Health & Safety (H&S) department within organizations operating in high-risk workplace environments, such as manufacturing plants, construction sites, and industrial operations. This stakeholder group is responsible for ensuring workplace safety, disseminating safety-related information, managing incident reporting, and maintaining compliance with organizational safety procedures and regulations.

The need for WorkBeacon was identified through discussions conducted at Fulkrum Interactive Technology with Nur Hazimah, the HR Lead who also oversees Health and Safety matters. Acting as a key stakeholder and working closely with the Product Owner, Nur Hazimah provided direct insights into the operational challenges faced by Health and Safety teams when supporting client organizations operating in hazardous environments.

From a stakeholder perspective, one of the main challenges is the lack of effective real-time communication between management and on-ground staff. Existing practices often rely on fragmented communication channels such as general messaging applications, phone calls, paper-based reports, or standalone systems that are not designed for real-time operational use. These approaches make it difficult for HR and H&S teams to broadcast urgent safety alerts promptly and consistently to all relevant staff.

Another significant issue faced by the stakeholder is the absence of acknowledgment visibility and accountability. Without a centralized system, HR and H&S personnel are unable to confirm whether safety alerts have been received, read, and understood by staff members. This limitation increases safety risks, especially during emergencies, as management cannot accurately assess staff awareness or response status in real time.

Additionally, incident reporting and tracking processes are often inefficient and delayed. Manual or unstructured reporting methods slow down escalation and response, while the lack of centralized incident records makes it difficult to monitor resolution progress, conduct audits, or perform post-incident analysis. These challenges reduce the effectiveness of safety oversight during daily operations and limit the ability of stakeholders to proactively manage workplace risks.

To address these problems, WorkBeacon is proposed as a complementary mobile application developed alongside a client's main operational system. The proposed solution focuses on supporting the Health and Safety department by providing real-time alert broadcasting, acknowledgment tracking, and structured incident management through a centralized mobile platform. By enabling faster communication, improved visibility, and better documentation, WorkBeacon aims to enhance workplace safety, improve response times, and provide operational transparency for HR and H&S stakeholders involved in safety management .



Figure 1.0 Meeting with Nur Hazimah discussing about the solution.

1.2 User Roles and Responsibilities

The main users of the system are categorized into two roles:

- Admin users (HR / Management) who manage alerts, incidents, and monitoring activities.
- Staff users who receive alerts, acknowledge instructions, and report incidents from the field.

Each user group has distinct responsibilities and system interactions, which are reflected in the system design and feature prioritization.

2. Use Case Analysis

2.1 Actor Identification

The system functionality of WorkBeacon is modeled using a UML use-case diagram to clearly represent interactions between users and the system.

The diagram identifies two main actors: Admin and Staff.

Admins interact with the system to create and broadcast alerts, manage reported incidents, track staff acknowledgements, and review incident records.

Staff users interact with the system by receiving alerts, acknowledging safety instructions, reporting incidents, and viewing the status of their submissions.

The UML use-case diagram helps visualize system boundaries, clarifies role responsibilities, and ensures that all required interactions are captured before implementation.

2.2 UML Use Case Diagram

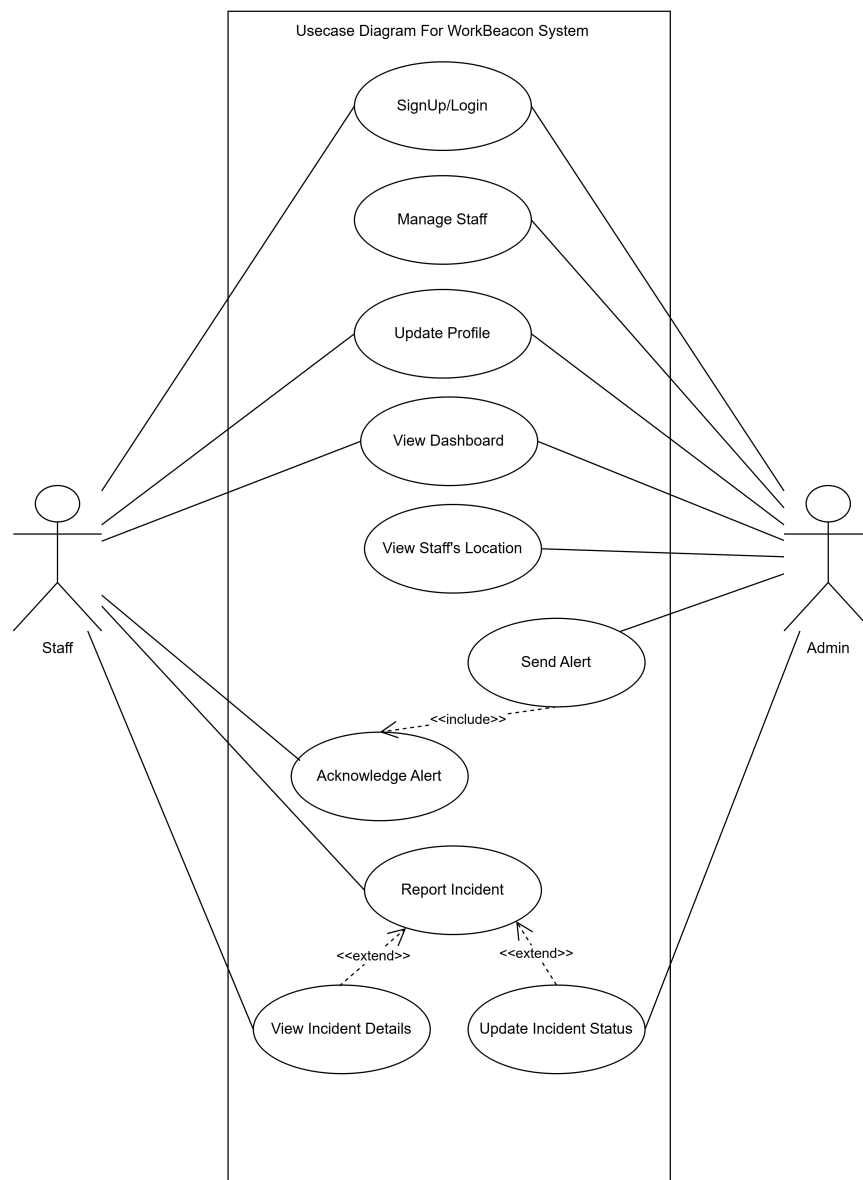


Figure 2.0 Use Case Diagram of WorkBeacon System

3. System Flow and Navigation Structure

3.1 Overall System Flow

The system flow describes how users navigate through the WorkBeacon application and how information moves between key screens.

For Admin users, the flow typically begins with authentication, followed by access to a dashboard displaying active alerts and incidents. From the dashboard, Admins can initiate alerts, view acknowledgment statuses, manage incidents, and access historical records.

For Staff users, the flow begins with login, followed by a simplified home screen that highlights active alerts and reporting options. Staff users can quickly acknowledge alerts, submit incident reports, and track updates related to their reports.

The navigation structure is designed to minimize steps during critical situations, ensuring that safety-related actions can be performed quickly and efficiently.

3.2 User Navigation Structure

The following is a navigation diagram for WorkBeacon App:

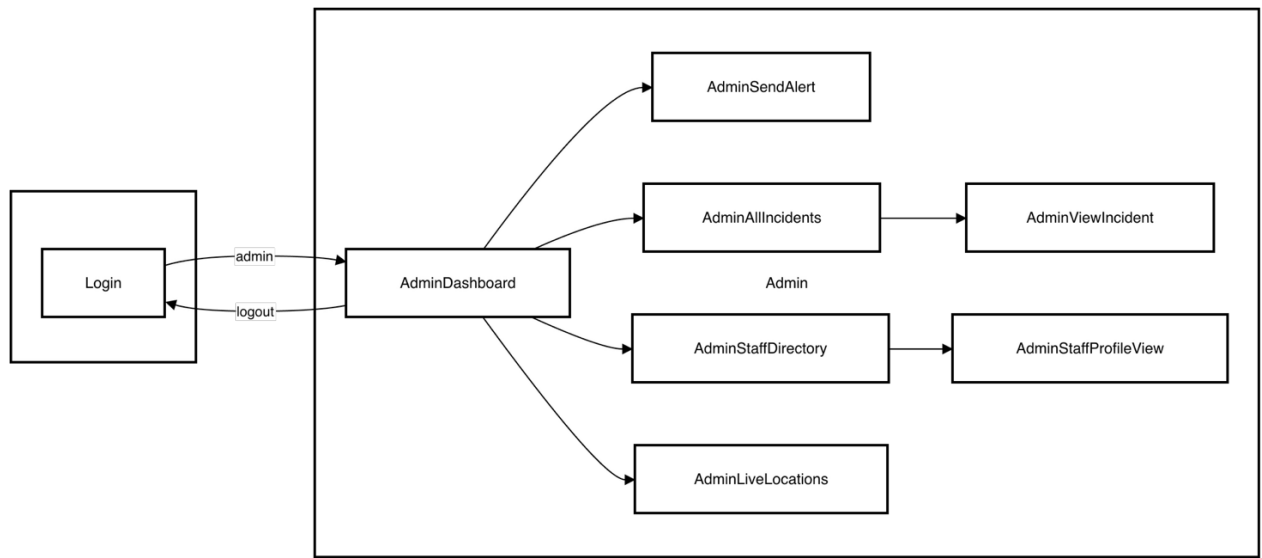


Figure 3.0 Admin Screen Navigation Flow

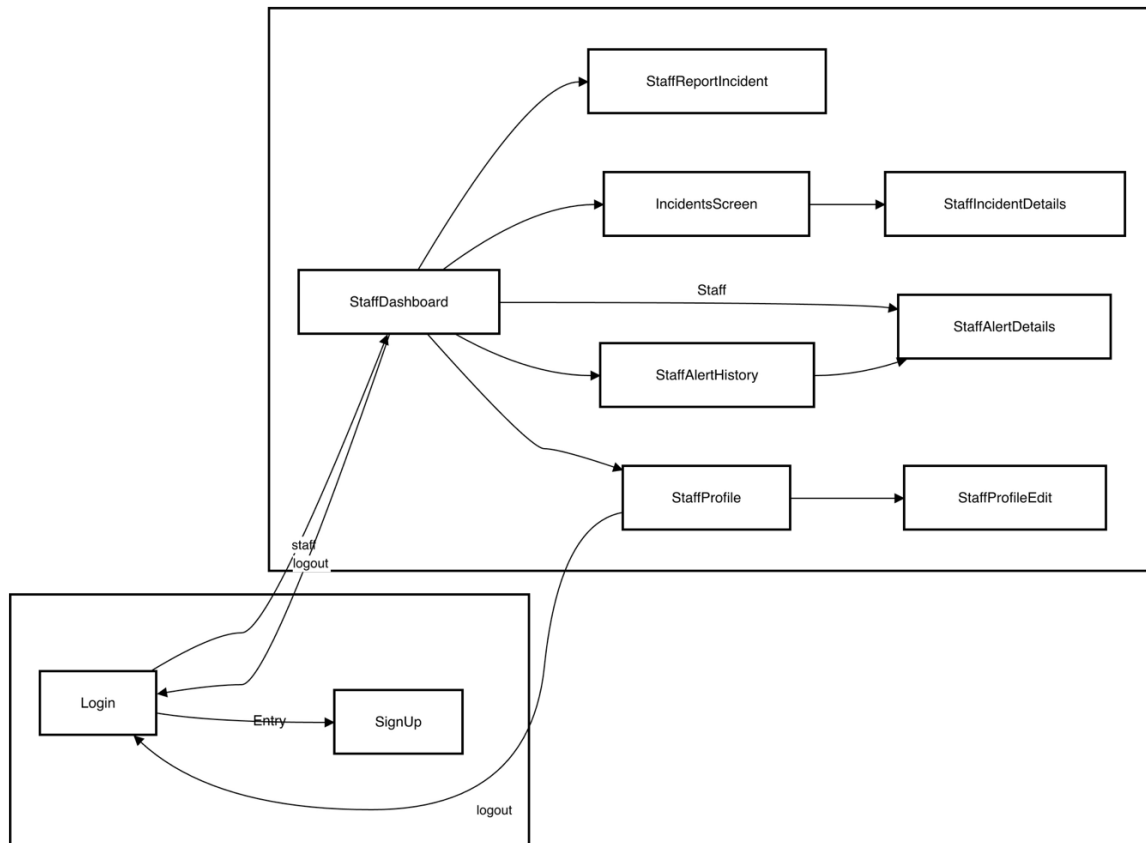


Figure 3.1 Staff Screen Navigation Flow

4. User Interface Design

User Interface Design is crucial for a good usability for an app. For WorkBeacon, our team have prepared a low fidelity designs that are used as a tool to elicit requirements from the stakeholder. Once confirmed, they were transformed into a high fidelity design close to what the user will use once the app has been developed. These designs were created to ensure that the client expectation is met.

4.1 Low-Fidelity UI Design

Low-fidelity UI designs were created to explore layout structure, content hierarchy, and user flow without focusing on visual styling. These wireframes emphasize functionality and usability, allowing early validation of navigation and screen layout.

Key screens such as the login page, dashboard, alert view, and incident reporting form were sketched to ensure that essential actions are easily accessible. The low-fidelity designs helped identify potential usability issues early and allowed rapid iteration based on stakeholder feedback.

These wireframes served as the foundation for developing more detailed and visually refined interfaces.

Below are the wireframe sketches for Low-Fidelity UI Design:

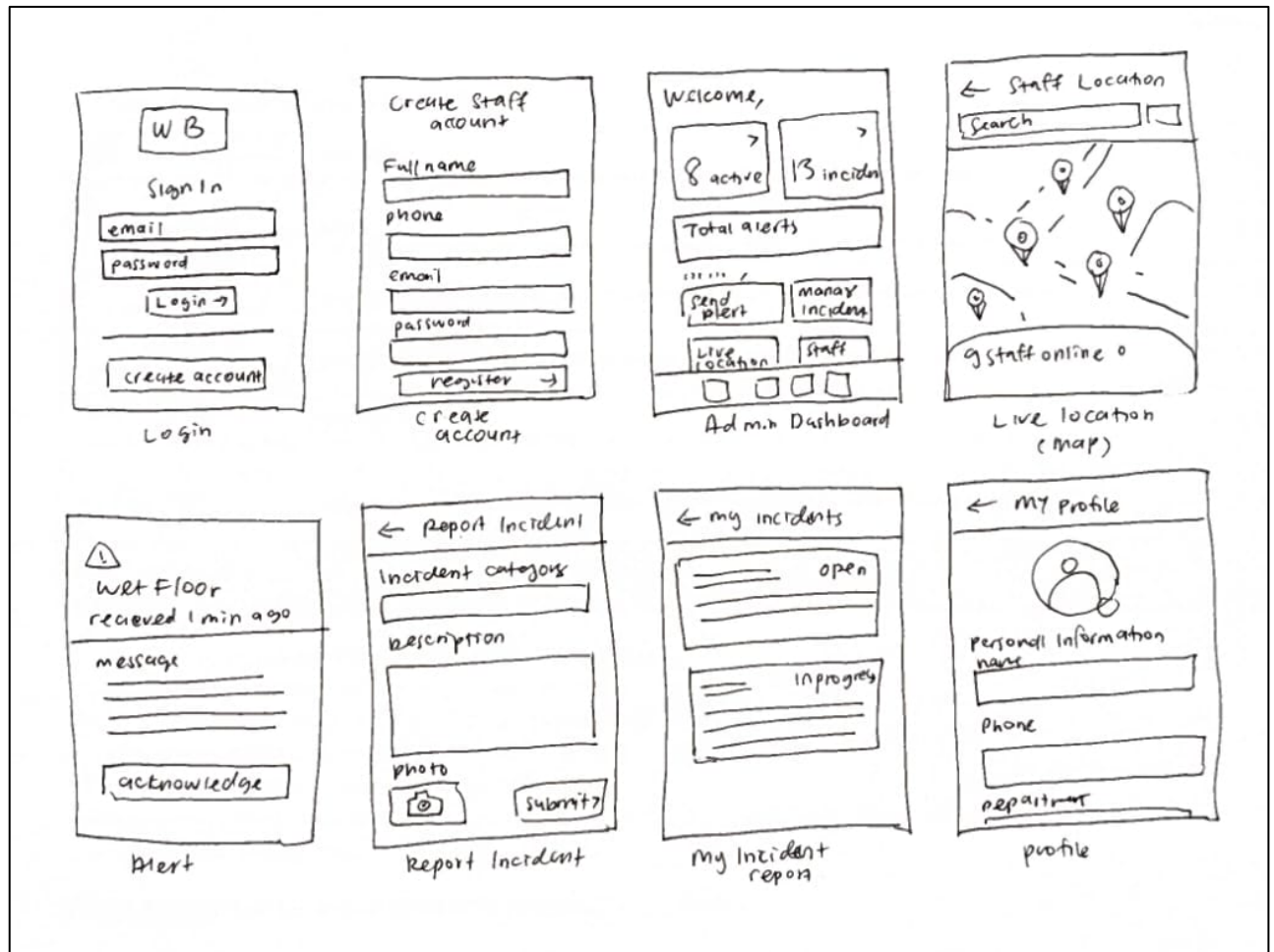


Figure 4.0 Wireframe sketches of WorkBeacon App

4.2 High-Fidelity UI Design

High-fidelity UI designs were developed based on the validated low-fidelity wireframes. These designs incorporate visual elements such as color schemes, typography, icons, and spacing to closely represent the final application.

The high-fidelity designs aim to provide a clear and professional interface suitable for workplace environments. Visual emphasis is placed on alerts, warnings, and status indicators to support quick recognition during high-risk situations.

Consistency across screens ensures ease of use, while responsive layouts allow the application to be used effectively on different mobile devices.

The following are the High-Fidelity UI Design screens across WorkBeacon App:

Screen Title	User	Design
--------------	------	--------

Login

Admin

W

WorkBeacon
Workplace Safety & Communication

Sign In

Email Address

you@company.com

Password

Forgot Password?

Login →

New to WorkBeacon?

Create Staff Account

Demo Credentials:
Staff: staff@workbeacon.com
Admin: admin@workbeacon.com
Password: any

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Dashboard

Admin

Welcome back,
WorkBeacon Admin Dashboard

Quick Stats

8

Active Alerts

13

Open Incidents

Pending Acknowledgments

Awaiting staff response

24

Quick Actions

Send Alert

Broadcast message

Manage Incidents

Review & resolve

Live Locations

Track staff

Staff Directory

Manage team

Analytics & Reports

View insights

Dashboard

Incidents

Locations

Staff

Reports

Incidents

Admin

← All Incidents

🔍 Search incidents...

Open

In Progress

Closed



Workplace Hazard

Medium

WB-2024-090

👤 James Martinez 📍 Warehouse - Bay 7
🕒 2024-11-30 at 08:45 AM



Environmental Issue

Medium

WB-2024-087

👤 Lisa Anderson 📍 Office Wing C
🕒 2024-11-27 at 11:00 AM

Incidents
Detail

Admin



WB-2024-089

Low

Current Status

✅ Resolved

Incident Type

Security Concern

👤 Reported By

Sarah Johnson

📍 Location

Parking Lot B

📅 Date & Time

2024-11-28 at 05:30 PM

Description

Broken security camera noticed at entrance.

Assigned To

Security Team

Notes

☐ Camera has been replaced and is now operational.

- Security,
2 days ago

Live Location

Admin



Send Alert

Admin

← Send Alert

Alert Type *

Emergency

Safety

Info

Alert Title *

e.g., Building Evacuation Required

Description *

Provide detailed information about the alert...

Image (Optional)

Upload Image

Optional visual reference

Location (Optional)

e.g., Building A - Floor 3, Conference Room B

Send To *

All Staff
247 employees

Specific Department
Select below

Require Acknowledgment

Staff must acknowledge receipt of this alert

Send Alert to All Staff

Staff Directory

Admin

← Staff Directory

Q Search staff...

All (8)

Active (6)

Pending (1)

Inactive (1)

- SJ

Sarah Johnson

WB-12001 •
Operations@workbeacon.com

Active
- MC

Michael Chen

WB-10001 •
Administration@workbeacon.com

Active
- EW

Emma Wilson

WB-12034 •
Warehouse@workbeacon.com

Active
- JM

James Martinez

WB-12056 •
Warehouse@workbeacon.com

Active
- LA

Lisa Anderson

WB-12078 •
Facilities@workbeacon.com

Active
- DL

David Lee

WB-12089 •
Maintenance@workbeacon.com

Pending
- JB

Jennifer Brown

WB-11045 •
HR@workbeacon.com

Active
- RT

Robert Taylor

WB-12012 •
Security@workbeacon.com

Inactive

Sign Up

Staff

← Back to Login

W

Create Staff Account

Join the WorkBeacon platform



Upload Profile Photo

Full Name *

John Doe

Phone Number *

(555) 123-4567

Email Address *

you@company.com

Password *

Confirm Password *

I agree to the Terms of Service and Privacy Policy. I understand that my account requires HR approval before activation.

Register

Already have an account? Sign In

Dashboard

Staff

Due to fire

Welcome back,
Hi, Sarah
Johnson

Active Alerts
1

My Reports
3

Latest Alerts

View All

⚠

Building Evacuation Required

Due to a fire alarm activation in Building A, all staff must evacuate immediately. Please

🕒 5 min ago

⚠

Wet Floor - Warehouse Bay 7

Spilled liquid creating slip hazard near loading dock. Area has been cordoned off. Please use

🕒 1 hour ago

Quick Actions

📷 Report Incident

📄

My Incidents

👤

Profile

Alert History

Staff

🔔 Alert History

All

Emergency

Safety

Info

Building Evacuation Required

Due to a fire alarm activation in Building A, all staff must evacuate immediately. Please proceed to

🕒 5 min ago

Wet Floor - Warehouse Bay 7

Spilled liquid creating slip hazard near loading dock. Area has been cordoned off. Please

🕒 1 hour ago

System Maintenance Tonight

The WorkBeacon system will be undergoing scheduled maintenance tonight from 10 PM to

🕒 3 hours ago

Alert Details	Staff	EMERGENCY Building Evacuation Required Received 5 min ago Alert Message Due to a fire alarm activation in Building A, all staff must evacuate immediately. Please proceed to the designated assembly point in Parking Lot C. Location Building A Acknowledge Alert
Incident Reports	Staff	← My Incident Reports Equipment Malfunction WB-2024-091 Conveyor belt motor making unusual noises and showing signs of overheating. The issue Building A - Floor 32024-11-30 Safety Concern WB-2024-078 Broken handrail on stairwell between floors 2 and 3. Poses fall risk. Stairwell B2024-11-15 Environmental Issue WB-2024-045 Water leak from ceiling in storage room causing damage to inventory. Storage Room - Floor 12024-10-22

Edit Profile	Staff	← My Profile SJ Sarah Johnson Operations Manager Staff ID: WB-12001 Personal Information Full Name Sarah Johnson Phone Number (555) 123-4567 Department Save Changes X Cancel
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5. Design Alignment with User Needs

5.1 User Needs Analysis

The overall design of WorkBeacon is driven by the needs of its primary users operating in high-risk workplace environments. The interface prioritizes clarity, speed, and accessibility to ensure that critical information can be acted upon immediately.

For Admin users, dashboards and management screens are designed to provide visibility over incidents and staff responses, supporting informed decision-making. For Staff users, the design minimizes complexity by focusing on essential actions such as alert acknowledgment and incident reporting.

5.2 Design Rationale and Justification

By aligning system features and UI design with real user workflows, WorkBeacon supports improved communication, faster response times, and enhanced workplace safety outcomes.

Table 5.0 shows the design rationale and justification UI-by-UI

UI / Screen	Target User	Key UI Elements	Rationale & Justification (Aligned to User Needs)
Login & Role Routing	Admin, Staff	Email/ID + password, role-based routing, forgot password	Ensures secure authentication and correct access based on role.

			Reduces confusion by sending users directly to relevant dashboards (Admin vs Staff) and supports role-based access control.
Staff Home (Alerts + Quick Actions)	Staff	Active alert banner, “Acknowledge” button, “Report Incident” CTA, shortcuts	Staff in high-risk environments need fast access to critical actions. Home screen prioritizes real-time alerts and reduces steps to acknowledge or report incidents.
Admin Home / Dashboard	Admin	Summary cards (Active Alerts, Unresolved Incidents), quick create alert, recent activity	Admins require operational visibility at a glance. Dashboard supports fast monitoring and decision-making (who needs attention now, what incidents are pending).
Alert Broadcast (Create Alert)	Admin	Alert type (Urgent/Info), message, target group/area, attach location/time, send	Solves difficulty in broadcasting urgent alerts by enabling structured alert creation and targeted distribution to relevant staff rather than fragmented messaging apps.
Alert Details + Acknowledgement List	Admin	Alert content, sent time, list of staff statuses (Acknowledged/Pending), filter/search	Addresses lack of visibility by showing real-time acknowledgement tracking, enabling Admin to follow up with non-responders and improve accountability.
Alert View (Staff)	Staff	Large alert header, readable message, safety instruction checklist (optional), acknowledge CTA	Prioritizes clarity during emergencies and supports quick acknowledgement with minimal interaction. High-contrast emphasis reduces missed critical information.
Incident Reporting Form	Staff	Incident type, description, location (auto/manual), photo attachment (optional), submit	Fixes inefficient incident reporting by providing structured submission. Encourages clear descriptions and consistent data for faster response and escalation.

Incident Confirmation / Submitted Screen	Staff	Submission summary, reference ID, “Track Status” CTA	Builds user confidence that report was successfully captured. Supports user need to track the status of submitted incidents and reduces repeat reporting.
Incident List (Staff)	Staff	My incidents list, status tags (New/In Review/Resolved), sort/filter	Keeps staff informed on progress and encourages trust in the system. Supports need for incident status tracking without requiring manual follow-up calls/messages.
Incident Management Dashboard	Admin	Incident table, status filters, priority/severity, assignment, timeline	Provides centralized visibility for H&S teams to view, manage, and resolve incidents. Directly solves the absence of centralized tracking and improves response workflow.
Incident Detail (Admin)	Admin	Reporter info, incident description, attachments, location map, status updates, comments/log	Supports thorough handling and accountability from reporting to resolution. Enables structured escalation, documentation, and audit readiness (activity logs).
Status Update / Resolution Workflow	Admin	Change status, add notes, assign person/team, close incident, resolution summary	Ensures incidents are tracked end-to-end with documentation. Supports compliance and post-incident review by recording actions taken and resolution outcomes.
Activity Log / Audit Trail	Admin	Chronological log of alerts/incidents actions, who changed what & when	Supports auditing, compliance checks, and safety performance review by maintaining traceability of actions, improving operational transparency.
Staff Live Location (Map View)	Admin	Map with staff pins, last updated time, search by name/team, geofence areas (optional)	Addresses Admin need to track staff live locations for coordination during emergencies and operational monitoring in high-risk zones. Shows freshness to avoid relying on outdated location data.

Notifications Center	Admin, Staff	List of alerts + incident updates, unread indicators, quick actions	Ensures users don't miss critical messages and supports real-time operation. Centralizes updates rather than spreading across multiple channels.
User & Role Management	Admin	Create users, assign roles, deactivate, reset password	Ensures only authorized personnel can broadcast alerts or manage incidents, strengthening security and preventing misuse in safety-critical contexts.
Reports / Incident History	Admin	Basic charts/tables, export (optional), date filters	Supports stakeholder need for centralized records for reporting and review. Enables analysis of incident frequency, response performance, and compliance documentation.

Conclusion

This Final Project Report has presented the design and analysis of WorkBeacon, a mobile application developed to support workplace communication and safety management in high-risk operational environments. The project was initiated based on real stakeholder input from the Human Resources and Health & Safety domain, where existing communication and incident reporting processes were found to be fragmented, inefficient, and lacking real-time visibility.

Throughout this report, key stakeholders and users were clearly identified, and their operational challenges were translated into structured system requirements. The use-case analysis and system flow design illustrated how different user roles interact with the system to broadcast alerts, acknowledge safety instructions, and manage workplace incidents. These design artifacts ensured that the system functionality aligns closely with actual user workflows.

Both low-fidelity and high-fidelity user interface designs were developed to demonstrate a user-centered design approach. The low-fidelity designs enabled early validation of navigation structure and usability, while the high-fidelity designs refined the visual presentation and interaction details suitable for real-world deployment. Each user interface component was justified based on user needs such as rapid response, clarity during emergencies, accountability, and operational transparency.

Overall, the WorkBeacon design addresses the core problems faced by Health and Safety stakeholders by providing a centralized, real-time mobile solution for alert broadcasting, acknowledgment tracking, and incident management. The project demonstrates the effective application of software design principles, UML modeling,

and user-centered interface design in solving practical workplace safety challenges. As a foundation, this design can be further extended and implemented to enhance safety operations and support future system enhancements.